

ActInf Livestream #039.0 ~ "Morphogenesis as Bayesian inference"

Livestream | 1:23:38

[Music]

[Music]

all right hello and welcome everyone to
active lab live stream number 39.0

it's february 25th 2022

welcome to the active lab we are a
participatory online lab that is
communicating learning and practicing
applied active inference you can find us
at the links here on the slide

this is a recorded in an archived live
stream so please provide us with
feedback so we can improve our work
all backgrounds and perspectives are
welcome and we'll be following good
video etiquette for live streams
if you want to learn more about
coms which is the organizational unit
that puts on all these live streams and
many other products or any of the other
activities in the active lab go to
activeinference.org

all right well

today in active stream number 39.0 the
goal is to learn and discuss

this really awesome paper called
morphogenesis as bayesian inference
a variational approach to pattern
formation and control in complex
biological systems it's by franz
koochling carl firsten georgiegiv and
michael levin from 2020

and just like all the dot zero videos
and indeed all our videos it's just an
introduction to some of these ideas it's

not a review
or a final word
that being said we're going to go over
the aims claims abstract roadmap keyword
etc of the paper and then
with a focus on the first parts of the
paper and less so on the formalism and
more on the big ideas we're going to go
over the paper and the figures
and that will put us in a good position
cognitively or morphologically we might
even say for
the discussions in the coming weeks 39.1
and 39.2
when we'll get to unpack this so if you
would like to it'd be awesome to have
you participate live or if it's past
39.2 you can still ask questions
all right we'll start with introductions
and warm-ups
we can each
say hi and maybe one thing that was
exciting or interesting about the paper
which all reserved for after you but i'm
daniel i'm a researcher in california
and dean thanks a ton for joining
uh thanks natalie i'm dean i'm here in
calgary
and
what really got me excited about this
was
uh the idea that they talk a little bit
later close to the end we may not even
talk about it today is
the uh role that epigenetics might play
in terms of how cells behave and
and what sorts of things we might be
able to turn turn in terms of

simply um what we see in biological form
into some sort of static
statistical prediction device so i'm
really excited about that awesome
yeah
it's a
complex paper about complex systems so
we'll see what we can get done today but
i'm really excited to bring
active inference into
this domain at least as far as our
discussions go into the morphological
and the morphogenetic areas
and it's just bringing like another
lens another way that we can apply
active inference as a filter
another set of phenomena that we're
bringing into
this active inference
realm
so we'll just go right into the big
question so i'll read this and then feel
free to give a thought
the big question or at least one way to
put one of the questions is how can we
find an integrative path between math
and morphogenesis morphogenesis means
form
like physical form morpho and genesis
meaning how it arises so how does
physical form arise and how is that
related to math
and to kind of unpack that here's some
quotes
the molecular mechanisms
which are like signaling pathways of all
different kinds
underlying development and regeneration

in biological organisms
have undergone a tremendous amount of
knowledge gain in the last several years
due to genomics and biotechnology
advances
so that's kind of the
one side with the biological and then
from the mathematical and the formal the
emerging field of the free energy
principle in pattern formation
not always morphogenetic patterns but
can be cognitive patterns rhythmic
patterns
provides an essential quantitative
formalism for understanding cellular
decision making in the context of
embryogenesis regeneration and cancer
suppression
so if that's the if
then
how might the mathematical formalism
reveal new understandings about
developmental change in evolution
and for designing new interventions in
regenerative medicine settings so what
do you think about that
well i just love anytime that two things
that seem disparate or distant are are
brought together because i think all
sorts of good things can happen and i'm
going to do a little shout out that's
kind of what i would like to
see if we can organize in august with a
couple of other papers that don't start
out necessarily appearing to be
close or proximal but by bringing
together all kinds of interesting things
can happen

awesome

okay

to the main aims and claims of the paper

so the paper is in physics of life

reviews 2020 and we already mentioned

the title and the authors

just to select a few aims that the

authors represent and then a couple of

key claims that are relevant for the

paper

so as far as aims go one of their aims

is to introduce an overarching concept

that can predict the emergence of form

and the robust maintenance of complex

anatomy

another aim is to

derive the mathematics behind bayesian

inference and use simulations to show

simulations that are consistent and

within the bayesian framework to show

that the formalism can reproduce

experimental top-down manipulations of

complex morphogenesis

and then just a few of their claims more

of which we'll be unpacking in the

coming minutes

classic

i.e dynamical systems and analytical

mechanics

approaches such as least action

principles

are difficult to use when characterizing

open far from equilibrium systems that

predominate in biology so that's kind of

a big challenge that's come up again and

again which is we might have really

great equations for a pendulum maybe

even a pendulum with friction but how

about an active pendulum that's doing
strategy and trying to run away from you
a bit harder to put those same equations
on
the bayesian inference framework treats
cells as
information processing agents
where the driving force behind
morphogenesis is the maximization of a
cell's model evidence
which is to say its reduction of
uncertainty relative to a generative
model which is what we're going to be
unpacking
and
we had previously read this
about how they're applying free energy
principle to morphogenesis
when it goes well
like during embryogenesis
and results in a zygote becoming like an
adult life form as well as all of the
ways that it's maintained during that
stationarity of healthy life and
including cases where it might be
relevant to even think about making
interventions like medical settings
so those are a few of the aims and
claims
anything to add on it
yeah just real quick it's really
interesting that they said top down and
then we're going to see some
slides where there's a top top
and
a down down so
stick around
because there's actually going to be

something kind of interesting happening

here

okay

how about you read the first half of the

um the top half of this abstract slide

yeah sure recent advances in molecular

biology such as gene editing bioelectric

recording and manipulation and live cell

microsc

microscopy

using fluorescent reporters especially

with the advent of light controlled

protein activation through optogenetics

have provided the tools to measure and

manipulate molecular signaling pathways

with unprecedented spatial temporal

precision this has produced

ever-increasing detail about the

molecular mechanisms underlying

development and regeneration bought in

biological organisms however

an overarching concept

that can predict the emergence

of form and the robust maintenance of

complex anatomy is largely missing in

the field

classic

i.e dynamic systems and analytical

mechanics approaches such

at least action as least action

principles are difficult to use when

characterizing open

far from equilibrium systems that

predominate in biology similar issues

arise in neuroscience when trying to

understand neuronal dynamics from first

principles so they're basically covering

their

bases here by making sure that we understand this is open system yes increases in tools in biological systems like genomics microscopy optogenetics are helping us reduce our uncertainty about the what but without a bit more information about the how and the why etc that's the overarching concept that's largely missing so we can grind up the cells and sequence the genome we can extract the rna and we can even tell what the ratio of different rna molecules is but it turns out that that type of information alone doesn't help us understand why two related species or the same species in two different contexts or the same genome in two different tissues or the same tissue in two different environments has such different morphology okay and then the second half in this neurobiology setting a variational free energy principle has emerged based upon a formulation of self-organization in terms of active bayesian inference the free energy principle has recently been applied to biological self-organization beyond the neurosciences for biological processes that underwrite development or regeneration the bayesian inference framework treats

cells as information processing agents
where the driving force behind
morphogenesis is the maximization of a
cell's model evidence
this is realized by the appropriate
expression of receptors and other
signals that correspond to the cell's
internal ie generative
model of what type of receptors and
other signals it should express
the emerging field of the free energy
principle in pattern formation provides
an essential quantitative formalism for
understanding cellular decision making
in the context of embryogenesis
regeneration and cancer suppression
okay and then if you read the first
um two parts here on the second slide
yeah no problem in this paper we derive
the mathematics behind
bayesian inference as understood in this
framework and use simulations to show
that formalism can reproduce
interesting experimental top-down
manipulations of complex morphogenesis
first we illustrate the first principle
approach to morphogenesis through
simulated alterations of anterior
posterior
axial polarity
the induction of two heads or two tails
ah spoiler alert as a planarian
regeneration
don't you hate it when the abstract
spoils the paper
then we consider aberrant signaling and
functional behavior of a single cell
within a cellular ensemble as a first

step in carcinogenesis the formation of
cancer

as false beliefs about what a cell
should sense and do

inference perception action inference

we further show that simple

modifications of the inference process

can cause

and rescue

mispatterning of developmental and

regenerative events without changing the

implicit generative model of a cell as

specified for example

by its dna

this formalism offers a new roadmap

for understanding developmental change

in evolution and for designing new

interventions in regenerative medicine

settings

speaking of the new roadmap

let's look at the roadmap

kind of cool that they actually

say that it is

providing a road map so we've talked

about guides

we've talked about road maps where do

you see all of that before we look at

the section titles

well

a guide i think

basically gives

sort of gives you um

the parameters i think a roadmap is a

lot more specific in terms of the

sequencing but that's just my

interpretation of it i think

other people can interpret roadmap

roadmap differently in terms of sort of

the relational aspect of how the
different parts come together and we've
talked about that in other live streams
but

i don't want to get bogged down in this
part right now because

i think we got a lot of stuff to cover
here

cool yeah so the paper begins with an
introduction to bayesian inference
section 2 is mathematical foundations
that are building on bayesian inference
so going from kind of bayes theorem to
several dynamical systems
representations some of which we've
discussed before others which will push
it into a little bit of a new area

then

the key

contribution and the focus of the paper
is modeling morphogenesis

and so instead of just presenting bays
and then variational bays and then free
energy minimization on

a bacterium like we saw axle constant do
in live stream number 34

we're going to take that
momentum

and move it towards an application of
modeling in morphogenesis

and that's where we'll talk about like
what model they construct and how it
relates to some of the formalisms that
were brought up earlier

and then

there's a discussion and conclusion
section which we won't go too much into
today in the dot zero but it's going to

be awesome to go into it in the dot one

and dot two

because there's a ton of awesome

content and material

all right

so here's the keywords that were listed

and we won't go into um

[Music]

most of them

just to say that they're going to come

up in this discussion we have free

energy principle bayesian inference

morphogenesis and developmental biology

regeneration

kind of got some mathematical stuff

we've seen before

we have a triad

of

developmental biology which broadly

understood includes embryogenesis as

well as

aging and cancer and all these other

developmental things that happened

during biology and top-down muddling and

maybe we'll see what that means coming

up

okay well to get right into it

section one is an introduction to

bayesian inference

so

here's a nice picture by sasha

and here are the opening lines of the

paper

they wrote and then feel free to give a

thought

evolutionary change results from

mutations in dna and selection acting on

functional bodies

thus it is essential to understand how
the hardware encoded by the genome
enables the behavioral plasticity of
cells that can cooperate to build and
repair complex anatomies
indeed most problems of biomedicine
repair of birth defects regeneration of
traumatic injury tumor reprogramming etc
could be addressed if prediction and
control could be gained over the
processes by which cells implement
dynamic pattern homeostasis
the fundamental knowledge gap and
opportunity of the next decades in the
biosciences

is to complement bottom-up molecular
understanding of mechanisms
with a top-down computational theory of
cellular decision making and info taxes
movements towards information
so

dean what do you think

what should we study and why and how is
this related to active and fep

okay well my background in genetics is
is near zero but what i did come away
from with that last bullet in particular
was

uh some one of the things that i
beat like a dead animal which is that
you need a minimum of two and i think
that the

the bottom up sort of building blocks
aspect of it with the top down how do we
get a better grip on what the signals
are that are passing between
all of those bricks um was really
interesting and so

carrying on through

i i kind of wanted to open that up and
see what that um that marriage looked
like

awesome

so speaking as someone who started in
the science journey with genetics pretty
much

it always was so interesting how
genetics would frame itself as if the
bottom-up information flow could just
simply replace theory we'll sequence so
many genomes we won't need to speculate
about evolution or we'll have your
genome so we won't even need to ask
about what medicine is best
and that's kind of this bottom-up
molecular understanding which is vital
it's essential it's important it's
changing a lot

but it's this bottom-up

sub-unit description

the what and the where of the subunits
and how they touch and all of that
and that is always met implicitly and
increasingly explicitly with a top-down
theory of cellular decision-making and
infotaxes and so we're always kind of
meeting in the middle with empirical wet
lab biology revealing these mechanisms
and these findings

and we're meeting that as researchers
doing model-based science with what are
being called here top-down modeling
top-down computational theories
okay that's just how they open up and
also set a challenge for the coming
decades

it's their expectation of the next few
decades
then they provide a math roadmap
which is an awesome paragraph
so they write we lay out the
mathematical foundations of a type of
bayesian modeling
that they're going to use to simulate
pattern formation
so red bayesian modeling
orange we start by identifying a
lyapunov function
that can be used to analyze and solve
any dynamic system using the fundamental
theorem of vector calculus i.e the
helmholtz decomposition
so we're using bayesian modeling and
layering in what we'll talk about soon
the leopard function
green we use it to characterize the
generalized flow of systemic states
states of the whole system in terms of
conversions to a non-equilibrium steady
state a ness
next layer blue
we then introduce the notion of a markov
blanket that separates the external and
internal states of the system
and finally purple
we can then replace the leopard function
with a variational free energy function
to solve for the evolution of internal
and active states
that's
of the particle
and thereby characterize
self-organization in far from
equilibrium systems that can be

partitioned into a cell
and an external milieu
that's the math that they go through
that's the math rainbow then
sections after that
apply this formalism to illustrate
morphogenesis and neoplasia which is the
growth of new forms neoplasia
using simulations
and what's so cool is this
trajectory
from 400 to 700 nanometers
from the bays
through the vector calculus and flow
states
to the finding or the figuring out of a
markov blanket
and then
the usage of the markov partitioning
legitimated
by the orange and the green steps
to find a heuristic way to go about
solving
what was
needing to be addressed right in the
beginning
we've seen some variants of this
trajectory multiple times
and then
in this paper they studied morphogenesis
and so then they model morphogenesis but
say for a couple of words
a different paper could have said we do
the base thing the lyapunov thing the
helmholtz
the nest the markov and then we're back
to doing far from equilibrium
strategy in

online teams

and then we applied it to online teams

and so

let's keep in mind like the paper can be

long there are a lot of equations but

where is this broader trajectory that's

helping us fine-tune our regime of

attention and update our generative

model about what we are doing here and

how it's different than other approaches

and then where is the part that's

actually studying morphogenesis

just things to keep in mind okay

anything to add on this

well you

correct me if i'm wrong on this but i

think what we're doing is

essentially

taking

as you said we can we can look at a

variety of things we can look at

the physical nature of ants we can look

at cells we can look at

online teams

whatever we're looking at what they

basically said is

there is a there is a parallel

way of looking at the formalisms

that exist in those contexts with that

with that

type of content

that's essentially what this said to me

there's a way of bumping it up into that

sort of statistical mathematical realm

and now we're going to walk you through

what that is

great

okay so into

the bay

the bay's area where i grew up

so

sitting by the dock of the base all

right oh i thought so i'm gonna say

where the water well where the

watermelons grow carry on

base theorem rests on three basic axioms

of probability theory and is used to

relate

the conditional probability of an

unobservable event a okay so the

vertical bar means conditioned upon or

given

and so what we really want to know is

like the probability of something that

we cannot observe

conditioned on what we did observe be

so you're never observing the lightning

strike you're observing photons on your

receptor and then you want to know about

whether there was lightning in that

cloud over a few miles away so we want

to know the conditional probability of

an unobservable event a that's this left

hand side

we want to relate that to

an observable quantity b

that's what the data are

and that's going to be calculated using

the bayes theorem as

the probability of the data

given

the unobservable state which is called a

prior so that's like

had to do with cats and meowing or in a

previous example or it has to do with

like

if there were lightning yes or no what
would be the likelihood of me observing
photons
multiplied by the probability of those
unobservable states happening which is
of course a prior about them but it
could be
set with empirical data
divided by how likely the data are which
is called $p(b)$ the marginal likelihood
or evidence
okay so $p(a|b)$ is called the
posterior because it's like after
you've done your inference you want the
correct posterior which is like updated
beliefs about how likely the thing that
you care about a is given what has come
in
priors are like prior to this round of
inference
what are your beliefs about how likely a
is
and then
the evidence coming in is b
okay
and the likelihood is how likely the
data are
under the prior distribution
so that's bayes theorem and we've talked
about it multiple times and i'm sure
there's other awesome videos and courses
that can explain it even better
how are they using it here
to describe the dynamics of an ensemble
of information processing agent like
cells
as a process of bayesian belief updating
we need to relate the stochastic

differential equations governing
newtonian motion
and biochemical activity to the
probabilistic quantities above
long sentence but
it's kind of saying we need to connect
these variables which are informational
they're probabilistic they're
statistical to
movement in space
movement in a newtonian coordinate
system
because we need these variables to
measure spatial movement so we can study
morphogenesis this is fairly
straightforward to do if we
associate biophysical states with the
parameters of a probability density
and ensure their dynamics perform a
gradient flow on a quantity called
variational free energy
variational free energy is a quantity in
bayesian statistics that
when minimized
ensures the parameterized density
converges to the posterior belief as we
will see below
so yes this is our first slide on bayes
theorem and we did just kind of go from
the total
description of the variables and what is
bayes theorem out of first pass to this
introduction of the variational free
energy so
it can be accorded out and unpacked a
lot more but
what variational free energy is doing
is helping us tractably

converge to
the appropriate posterior
belief and that's its role here
okay
anything to add on this one dean
well i'll ask you a question so if you
were to talk to this to somebody for the
first time
not people who probably listened to
other of our live streams
would you say that it would be
um dangerous to say that we're taking
something that's kind of ratio based
and now
using that to try to build a model of
what our predictions
are going to
be based on
is that
is that fair to say
i believe so
it's like
one way to approach that
i hope this would be accessible the
ratio-based part is on the right-hand
side
and um one could say depending on like
how much data you're getting and what
you're trying to estimate it could be
very hard to go from
observations to inference on hidden
states
pretty fair conclusion
so
what we could do
would be construct a smooth
approximatable model
so that when we slide to the bottom of

our model just like the ball sliding
down the hill
we would actually be doing the same
thing as
fitting this effectively
so instead of like number crunching and
putting everything into this equation
and running the calculation
we're going to construct
a bowl
which is the variational free energy
that we can do gradient descent on that
is going to converge to a really good
solution
do we construct it or do they do they
kind of tell us that
we should assume it if we want to make
it tractable
assume the bowl
when we pick up this instrument
we're making variational free energy
bowls
yeah okay good
and we've seen this before
um
in
live streams
26
32 and 34
we did some variants on moving from
bayes theorem and exact base
to
variational inference which uses this
variational free energy value
so those are a few times that we've seen
this discussed before
in neuroscience the minimization of
variational free energy

is referred to as

active inference

we will see that the basic when the

basic condition for an inference type

description of a system

namely the existence of a markov blanket

separating external and internal states

is satisfied

agents such as biological cells form

into organized conglomerations based on

their generative model of how their

blanket states influence

and are influenced by

external states in the external milieu

so the connection between bayes theorem

and markov blankets is the unobservables

are external states

the observables are blanket states

it's a pretty fair model

whether it also provides like strong

guarantees on optimal perception or

optional action that's another

discussion but

at the very least it's a fair framing

that external states are unobserved if

you could observe them they'd be

observables then they'd be more like b

than a

so we're talking about a situation where

we have some a's and we have some b's

and we're going to think about that in

terms of a blanket formalism

and

the dynamics of a system with a markov

blanket that self-organizes to

non-equilibrium steady states can be

described as a gradient flow on this

computable variational free energy bound

so we can set up the blanket and it
turns out that we don't have to do the
ratio number crunch we can do
the bowl gliding down
that's the beginning part on just bays
that takes us from bayes theorem to
introducing variational free energy and
doing gradient descent on variational
free energy
so that one can effectively essentially
perform bayesian inference
cool
perfect okay section two mathematical
foundations
all right
so section 2.1 stability and convergence
in coupled dynamical systems and
subsection 1 the helmholtz decomposition
so they write the helmholtz
decomposition states that any
sufficiently smooth
i.e possessing continuous derivatives
vector field f
can be decomposed into an irrotational
curl free
so kind of
straight up and down putting the ruler
and finding the slope of a hill
and
a solenoidal divergence-free vector
field the divergence free solenoidal is
like putting the ruler
at ground level
and then being able to make an iso
contour around a hill
because an irrotational vector field has
only a scalar potential
and the solenoidal vector field has only

a vector potential we can express the
vector field as
insert formalism two
and just so people know the upside down
triangle is called del or nabla which
means harp due to the shape
so del
phi
is
the irrotational
vector field
del a is the solenoidal field
we talked about this mostly in
um
number 26 and 32
where we talked about the partition or
the decomposition
into a solenoidal and a gradient
as well as this housekeeping term which
was
everyone's favorite housekeeping term so
we're not um
having a housekeeping term here
but we're keeping in mind that it's not
implausible that
it's
still there
it's just that there was a whole paper
coming out after
this morphogenesis paper
that
talked about the housekeeping etc so
again it wouldn't be surprising if a
future morphogenesis paper or project
did have this housekeeping term but um
32 was actually a later paper so that
might be a more advanced placed um or
updated place to check out

how the partition i'm sorry the um
helmholtz decomposition works
and here was a image from
just showing how you could have like a
kind of
um
hill
and in physics you could have hill
climbing and trying to get to the top
or
sometimes it's easier to think about
that as like the flip
and the ball rolling to the bottom of
the hill so whether it's the optimistic
biologists talking about going up to the
fitness peak or whether it's the
pessimistic physicists talking about
going to the bottom of the energy well
it's the same idea
that there's a particle on a landscape
and there's been a ton of work on
passive particles that's not to say
immobile particles but particles that
roll and get trapped and then once
they're trapped they stay or they jiggle
stochastically
we're studying active particles and so
that's where action selection comes into
play landscapes of energy with balls
rolling down them but they're kind of
like jumping beans where they also have
a few tricks that they can play
okay anything to add here
just just a reminder to people that
that's a density that we're looking at
not not a
not a
not a vacuum it's a

i know most people when they hear like
if i was to talk to younger people and
they started imagining a ball rolling
down a hill they would they wouldn't
necessarily

confirm that it's an
actual density that we're looking at so
just reinforce that

yeah like the bottom right here it's
kind of like you're looking down at an
ink drop

and that's where it's densest is where
it's darkest and then that is like a
landscape where where it's darkest is
highest

right

okay

section two one two the openiv functions
okay so the opposite functions have been
used extensively in dynamical systems
theory and engineering

to characterize the stability of fixed
points of a dynamical system

the operative functions are generally
defined for smooth systems through the
following conditions

so f is going to be the leopard if
function

and x is going to be a point

so

a fixed point is going to be x^*
and a non-fixed point is going to be
just regular x so x^* the special
points where if you are there you're
perfectly balanced

so

this is saying that
at fixed points

the Lyapunov function

evaluates as zero

and it's not zero

if it's not a fixed point

and

the Lyapunov function

change in L with respect to time

it's below zero

for all x

equation three a requires the Lyapunov

function to be minimal for fixed points

x^* representing local minima

and b denotes transitions to these fixed

points over time so what does it mean to

have a Lyapunov value of 0

above zero or below zero so if you're at

zero just like it said with the fixed

points that means that it's not moving

at all

a Lyapunov exponent below zero means

that things are converging towards

the origin

bigger than zero means diverging away

specifically what it means is two points

that are really really close to each

other

do they stay exactly the same distance

away that's zero

a negatively operative means that two

points close to each other will converge

so that's a convergence attractor

and then positively open of exponents

the opposite values

mean that two points that are placed

very close to each other will diverge

away from each other

i hope i have not oversimplified or made

unforced errors about the opponent

functions because it's obviously a really technical area and they write following 16 we can generalize this local leopard function of stability

so zero equal fixed point positive equals divergence negative equals convergence to a globally open a function that plays the role of a potential function of any dynamical system

and we won't go into it but it is the citation to this ryoshi paper from 2013

and they do some pretty interesting math so if somebody's more familiar with this it would be awesome to hear about but what this is doing is it's taking this same dynamical systems perspective that we've been thinking about about things changing through time and then overlaying

on that landscape a stability landscape and so if we think about like this hill on the top right again hopefully not being

wrong or misleading here at the very top of the hill

it's like a fixed point it's an unstable fixed point but if one were perfectly balanced there we could just say um it could get knocked off by stochastic things which we'll come back later but it's perfectly balanced it's like a fixed point two points that were very close to each other

on a hill

two balls placed right next to each other would start to diverge as they rolled down the hill

so that would be like a slight divergence

and so that is what the lyapunov function as potential function is allowing it's allowing us to take the landscape of ball rolling where gravity is the potential function

the higher up the more potential you have and then it rolls down because the dissipation of potential energy

and they're going to

um

show that any dynamics with lyapunov function so any smooth differentiable etc landscape has a corresponding physical realization a friction force a lorentz force and a potential function

so we can think about this leopard function being related to potential functions like something that draws the system back

that's where the negative lyapunov comes into play okay dean

just a quick thing daniel

when we were setting up the slides between those two bullets um they they talked about something called a saddle point did you understand what a saddle point was

previous to reading this or do you know what a saddle point is now

yes let us i'm just pulling up the um paper in the window okay so people can see it

um okay so

yeah the stability um okay

so it's the following sentence so we can

generalize the leap the function to a

global function that plays the role of a

potential function of any system

this follows by generalizing condition a

to allow for saddle points

so a saddle point is when the landscape

is shaped like a saddle

and there's like um

so if it were on top of a hill

the curvature is negative with respect

to all directions

right in a saddle point the curvature is

negative with respect to one of the

dimensions

but it's positive with respect to

another dimension

and so it's just like just like a saddle

yeah

okay

why do they call it that um and so

the ball on the top of the hill

is like an infinite multifurcator

like a little jiggle it could go any

direction

because it's unstable in every direction

saddle points

induce a symmetry

where that ball is never going to roll

up the hill

it's going to fall to one side of the

saddle or the other side of the saddle

and one of those could be like um

a certain attractor and the other one

could be like another elevation

basically

um

though i don't know the details of how
the saddle point is related to
some of this lyapunov stuff

okay

all right so following

16

so this is they're showing that the
leopardive is equivalent to a potential
function
in physics a potential function ψ
can be constructed to describe the flow
of

or forces acting on
a particle through a potential energy
gradient

so the potential
energy function f is
 ∇

ψ

so here's just one way to state the big
idea of equations five through nine
which are not discussed here

so the idea is create a potential
function

and when you think of potential function
think it's like kinetic energy
based upon elevation

but it's not for gravity and elevation
it's like balls falling down tall
buildings

but for

not gravity

for something that's related to
dynamical conversions

so if you were like at the bottom of the
bowl you'd only have a little bit of
potential energy

the potential function would be low if
you were really far up the ball
wall you'd have a high potential
function you'd have a long way to fall
that
is going to be reflected by a balance of
forces
namely an attractor force
which is the combination of the lorenz
force and the potential energy-induced
force
and then there's the dissipative force
and that's the frictional force
due to dissipative random fluctuations
so
that's like the velcro ball on the
velcro bowl
the friction is so high
that it still is being kept on the side
or it's rolling down very slowly or
something like that
combining these definitions we can
express the total balance total force as
a balance of the forces defined above
resulting in
and so just like if it was two colliding
vectors and then we were gonna express
the outcome resultant as like the total
forces being applied on something
it's like that but it's not about the
physical forces in the gravity it's like
more about
these forces
abstractly and so equation 10 is
expressing this balance of forces
on
x using a gradient landscape so there's
that

del psi

and here's the combination of forces

being expressed as

del psi dean

so

they use the word tensor here and when i

was reading this i didn't

i didn't see it i mean okay so if we're

having a tug of war on a rope

it might appear to the outsider upside

outside observer that there's a balance

but really what i read into this and

maybe i'm wrong was that there's a

tension

between these two forces

between the attractive force and the

dissipative force am i

am i misinterpreting that

someone has a different answer would be

awesome to hear it a tensor is like a

generalization of a matrix it's like a

matrix through time

i don't think that

it is directly related to tensile or

tensegrity

okay that being said you're right that

there is

a tension between these forces or there

can be

and that they like tension

can reflect

like a compromised position like if you

have a rope with tension on it

then the ending of the rope is going to

be like some compromise between the

various forces

and so we're thinking about like the

various forces

and then we're expressing it as like
this matrix
through time
that's what makes it a tensor not the
fact that the forces are in tension or
um opposition with each other yeah okay
i hope that's accurate okay
um
and then they follow 16 again
and transform that expression in 10
into a more standard form using the
diffusion tensor gamma
and a tensor q
this is like a rewriting
that helps them get to this format
where $f(x)$ is describing the flow of
states so that's the change in states
through time
and it's like a vector field
and it is going to be $q - \gamma$
so
tensor
minus diffusion
both of that so that multiplied by
that landscape
so that's kind of like the
decomposition
but if someone could explain this more
let's hear it
and my simple words this is where i i i
keep bringing up the idea that
maybe when we're looking at these things
through an active inference lens we're
not just looking through a frame
we're also looking through and they talk
about filters later on
so
so like

how do we how do we make sure that we're
including both
and that and sort of the tension between
both of those views working
together and being in tension with one
another so yeah
this is very this is very formal
but i think it kind of reinforces that
there's there's
what's inside the
the
the partitions and then what's going on
within them as well
yeah
and it um so it's describing the flow of
states resulting from this conservative
and dissipative forces so if q and γ
were the same
then the flow would be zero
if one is larger than the other then
it's going to be dominated by that
component with respect to being
multiplied by this landscape
so
we'll return to it many times
okay
variational free energy um
what did you want to add on this slide
well i i don't know that there any i
mean most people already know what we're
talking about in terms of variation free
energy but what they what they're doing
is introducing it in terms of so now
where does the model
the model of a of a
biological entity
where is that sourced from and that's
this sort of idea of these

these

these things like

posterior beliefs the foundation or the
the idea that we we don't walk into a
situation completely blind we have some
sort of prior

and we're modeling situations where
internal states can be interpreted and
framed as a generative model so it's not
they're focusing on the model aspect of
this as opposed to the generative
processing side they're gonna they're
going to talk about what the external
milieu is and where the generative
process is here shortly

great so

good point to pause the video and read
this because we're not going to read it
out here but you're absolutely right
they're focusing on the generative model
of the agent of the cell

not

doing that

analysis on the niche for example

all right

section 2.2.1

okay so this is a fun part

we can describe dynamics in generalized
coordinates of motion denoted with a
tilde

where x tilde is defined as

x tilde equals x that's like the

absolute state of x and then x with a

dot is the first derivative of x that's

called velocity

x with two dots is u or a two double

prime depending on the notation is the
acceleration

what people don't always know is that there are higher derivatives and they also have names so the third derivative is the jerk and then the fourth fifth and sixth are the snap crackle and pop so that's just kind of funny but the generalized coordinates of motion which came up in number 26 have to do with doing prediction on not just um how high or low something is or like on the number line left and right that would be like if x were just one number but even if it's just one number that we're tracking like body temperature for example we might be interested in kind of unpacking that into this generalized coordinate where we want to know body temperature and how fast it's changing and how fast that is changing and so one can imagine that once a zero has been achieved the higher derivatives are also zero if it's zero through time the higher derivatives are also zero if it's zero instantaneously that's not always the case but if it's zero through time then it's kind of like you've cashed out the derivatives the derivative of zero just stays at zero and so it's like if the position is unchanging the velocity is zero ends

if the position is five miles an hour
 going in the same direction the velocity
 is constant but the acceleration is zero
 so keep on carrying that out to how fast
 things are changing how fast they change
 et cetera et cetera
 and then they're going to take that $\tilde{x}$
 tilde
 how $\tilde{x}$ changes with time $\tilde{x}$
 dot
 equals
 the flow on tilde
 plus
 a variance
 so a noise or fluctuation a random
 fluctuation term
 this is the form of the Langevin
 equation
 generalized coordinates of motion
 plus
 a random fluctuation term
 um
 any of the
 mathematical colleagues would be awesome
 to hear
 how this is related to the Wiener
 assumptions
 etc but we're going to continue
 so
 using the Helmholtz decomposition we can
 now express steady state flow in terms
 of a divergence free component and a
 curl free descent on a scalar of the
 operative function I of $\tilde{x}$
 to obtain
 f of $\tilde{x}$
 equals there's the q minus γ that we
 saw a few slides ago multiplied by now

the I

so here we had f of x

not generalized coordinates of motion

q minus γ

Δ

ψ of x that was the potential function

now because the potential function and

Lyapunov are related to each other

we're looking at the generalized

coordinates of motion

using the same q minus γ Helmholtz

decomposition but multiplying it now by

the opening of exponents or I'm sorry

the Lyapunov function on

$\tilde{x}$

it's the solution at non-equilibrium

steady state and is exactly the same

solution

for

the flow of particles in the classical

treatment above

crucially we can now see the operative

function is the negative log probability

of finding the system in any generalized

state

the Legendre of $\tilde{x}$ is the negative

natural log of

how surprised

one should be p of $\tilde{x}$ this is known

as self-information of a state in

information theory surprisal or surprise

in Bayesian statistics it is the

negative log evidence so

the change in the generalized

coordinates of motion

f of $\tilde{x}$

breathe

is an irrotational solenoidal

decomposition

that's the left part of the right hand

side

multiplied by a lyapunov function

which also has the interpretation of

surprisal

so if you are exactly

resting so to speak

you'd be totally unsurprised

in summary any weekly mixing dynamical

system at non-equilibrium steady state

will events give us evidence for or look

like it is

a flow that can be decomposed into a

gradient flow on surprise and an

accompanying solenoidal flow

because we can associate the lyapunov

function in 18 with a free energy

the system is effectively minimizing

a free energy in its convergence to a

set of attracting states

which have a high probability of being

occupied

so it's a little

circular it's like

we're finding ourselves where we expect

to find ourself

and so if we just said well how

surprised are we

and let's gradient descend towards not

being surprised

just at a first approximation

we would find ourselves

in regions of low surprise

and we would be surprised to find

ourselves in regions of high surprise

is it is it circular or is it kind of

the byproduct of a recursive

process

again i'm asking i'm not questioning

it's a great question

in like a really

meta way all of math is circular once

the axioms have been stated because like

the equal sign is there so we're just

you can add one to both sides and it

would still be equal so is that circular

right um it's useful

and that's what matters for this paper

okay yeah

um

we're not going to talk about it too

much but in a dot one we absolutely will

and that's this least action principles

in 2-2-2

and they even give the example we love

to see

for example in colonies ants find the

path of least action to harvest food and

bring it to the colony

citation needed slash let's see that

but conceptually

no problem

this example considers their paths as

flow channels or trajectories finding

the least average action for each

instance of foraging given available

resources

so

what does least action principle mean

what does least action mean

and what is it necessary or sufficient

for because certainly foraging is not

the least biochemical expenditure of

energy

that's like the sort of

naive interpretation of least action
meaning just stay where you are
but we're thinking about a usage of
least
action that impels this ant to forage in
an adaptive way
such that even when it's foraging trip
is hard and the seat is heavy and it
gets like lost or something
that is still a realization of at least
action on something it's still a ball
rolling down a hill as efficiently as it
can
in a space
so it's a
nuanced topic that
it will be great to discuss
because they also write minimization of
action in an open system leads to
structure formation
so that'd be a good thing to unpack okay
carrying on
in dissipative random dynamical systems
action is not minimized for each element
of the system
but on average over an ensemble of
elements or repeated trajectories of the
same element
since self-organizing open systems are
not conservative
their structured flow is
quintessentially dissipative
good to
think about that and hear many
perspectives and questions on it
classically for a conservative system
the lagrangian is defined as L
this is

I not the operative now it's lagrangian
equals t minus v
so t is the kinetic energy
of the particles and v is the potential
energy

[Music]

and so that's how
the
trajectory of system states could be
solved for a conservative system
where the total energy is going to be
conserved

but

they're

opening up into this
dissipative and open
scenario

where this is not going to cut it
um

[Music]

a little bit more detail on least action
which we're not going to go into
okay

continuing on

two two three markov blanket

okay

a robust literature is developing about
around the ability of cells and other a
neural non-neural systems measuring
aspects of their environment via
specific sensors

they introduce here the markov partition
as this general case of
separation of states into four
categories

e

external states

sensory states s

active states a internal states i
and so the little e is like it's a
realization of big e that's the bigger
space
and then
 x tilde
comes back
and the generalized coordinates are
going to be
a realization of values of those
partitioned states
so we've talked about the markov blanket
before
but now we're interested in
the generalized
flow on
the blanket partition states
so
here's what the markov blanket looks
like
and
we'll talk more about it in the dot one
and dot two what do the arrows mean and
why are sense and action connected
what part of the cell are we looking at
is it a part of the cell is it a model
of a cell
all these things that have come up
before but just to say that's their
figure one and table one gives the math
formalisms
okay anything to add on markov before we
continue
no because we'll we'll unpack yeah
so
this is where they apply the markov
blanket
so m is describing the markov partition

that defines the underlying random
 dynamical system such the cell
 so that is like um
 the uh
 model that's the partition markov
 partition of the cell
 so inserting c into a and b is going to
 give
 there's
 a is actions and i is internal states
 so the top equation is basically saying
 the flow
 on action
 is a function of sense action and
 internal
 and then
 it has a right hand side so the flow on
 action
 the performance of action
 is
 and then there's that $\gamma - q$
 but now they have a subscript a because
 it's those about the action landscape
 multiplied by Δ on action
 $\tilde{\text{generalized action}}$
 and then a self-surprisal
 looking like
 the self-surprise of
 sense action and internal conditioned on
 m partition
 and the bottom equation in b 30 b is
 basically the same
 but it's the flow on internal states
 $f_{\text{sub } i}$ on
 generalized states of sense action and i
 and that same $\gamma - q$
 Δ self-surprise condition on model
 so we're seeing some patterns come up

and even if um it's like confusing and
we all definitely are confused by it to
some extent we're starting to see like
some patterns
why does it matter to focus so much on a
and i the action
and the internal states because those
are like the ones we control
we know we don't control external states
directly
maybe we can intervene so that they
change differently but we know we don't
control them
we also don't directly control sense
states we're getting dealt this hand
every second by the niche the generative
process
and so we can
take action so that we can expect
different sensory outcomes but even then
we wouldn't be controlling sensory
outcomes
and so
a lot of active inference comes down to
doing inference on
internal model
internal states
generative model learning
and
action action selection
flows on action
um
[Music]
this
bounds the surprise
on the particular states which is the
internal and the blanket states
through

control of the autonomous states which
is just the action and internal states
the states we control
because of the sparsity
of the blanket
not every node is connected to every
node
there might be a factorized and
tractable form
to bound our surprise
about the particular states
in general it would be very difficult
problem to solve
however we can replace the lagrangian
that's the one
here
that was like going to be used for
conservative systems but it didn't
really apply
um
with a variational free energy
functional or of a probabilistic model
of how a system thinks it should behave
or how we think the system should think
it should behave
so it's hard to solve these
generally but
in practice there are heuristics such
that these are approximatable
section 2.4 goes like one layer deeper
into kl divergence in the vfe but we're
not going to go into it
okay
we're just going to keep on plowing
through
two three is bayesian filtering and
self-organization
we're not going to talk about it now but

the big questions here are like how does
bayesian statistics relate to identity
and what is self-evidencing
and there's probably a lot of other good
questions we could ask like why does
bayesian filtering and self-organization
come here

all right

modeling morphogenesis so now we
actually get to the contribution of the
paper

um which is the modeling of
morphogenesis so they're going to
illustrate self-organization to ness
using the variational principles above
by trying to explain the behavior of a
model of pattern regulation by
consideration of information processing
and error minimization with respect to a
specific target morphology in this
setting the game changes subtly but
profoundly

it's a dean line if i ever read one

um

above

which is what we've reviewed
the dynamics of any random dynamical
system equipped with a mark called
blanket can be formulated in terms of a
gradient flow on variational free energy
that was just

here the flow on action and the flow on
internal states

here we turn this formulation on its
head

by specifying a generative model
an implicit vfe function
and simulate self-organization by

solving the equations of motion in
equation 34.

so they're going to specify the form of
the attracting set

v

uh the generative model

and then they're going to let it ride

so

we have to just talk down

yeah

they specify the external dynamics as
the generative process

that's the niche

and the generative model

of that process which is being described

by the flow of internal states okay so

we're going to look at the figures

and then we're not going to even like go

into the details so let's just look at

the figures and see what they're doing

so

here's some empirical biology happening

in the lab of mike levin and others

so

in a

it turns out that when you dissect out

the center of a flatworm a planarian

those cells will remodel into a new worm

so cutting out the middle and it reforms

into this

new

flatworm and red is the head and then

like blue is the tail

so

there's a self-organization

of this target morphology even from an

initially clumpy cell

what they're going to do in b

is showing like the final fixed point
of these different cells so here we have
like four different cell types there's
like the red neural or head cells then
yellow cells green cells and blue cell
and then basically like this like three
two one one
um and then the bottom one it's this one
maybe they change the green and the blue
like i think these two should be green
and the bottom should be blue but
minor point but it's an attractor
on

this target morphology

um

and then

c

is describing a little bit about how it
happens which is that cells are
constantly comparing their sensed signal
concentration like of a gradient of some
um morphogen

to expectations by minimizing their free
energy functional

so it's like what kind of a cell type am

i

and what should i be expecting
flip side what should i be surprised by
because to say what i expect is
the other side of the coin of what one
is surprised by okay so they're modeling
this empirical scenario where a flatworm
can regenerate its form from just a
clump of cells here

and that

if we think about the location of these
cells as being like a steady state
attractor like we want to have this body

form last
then if all the cells just get in line
like that everything's going to work out
all right
how do we take this
first jump that we made from the
empirical biology
to the
state space framing
here it is an x y state space
and now take the next jump
one more mathematical
here on the right side
is
that
x y positioning of each cell
this is reflected by the position
in
x and y
of the cells so like negative nine and
zero so that's like this top red one
zero is the midline so it's like y and x
and here like negative five it's a
little lower
and then like negative four and four
it's like one is on they're on the same
y elevation
but then one is four to the left and one
is um four to the right
and so this e star of x
is the position
so it's um a matrix of position because
there's like
um eight cells
and two location variables per cell so
it's like a two by eight matrix
and it changes through time so it's a
tensor

the e star c
is the external signals
and so
in this
steady state attractor
all three red cells are getting like
signal one here's the binary signal
there are four signals but they only
have an on or off state in reality
there's many many more signals than four
and there's a lot more uh nuance than
just on or off but it's a
it's a toy example all three red cells
are getting the exact same signaling
milieu
of one one zero zero so factor a and b
are diffusing near me
but not c and d
the yellow cells are experiencing
those two diffuses and one other one
the blue cells are experiencing the
first the fourth and then the green one
is another
and so
this is like the attractor state
for a stable
location
and signaling expression
so converging to this spatially makes
your body look like that but the way
that you get a beach ready planarian
body like that
is actually by reducing your
your surprise on signal expression
okay so here's them
running it through time
so it's a time-lapse movie montage
of simulations of morphogenesis

and so here it is converging towards on
the right side
the morphology that's been discussed
here
and so the cells start out with
undifferentiated they know what they're
sensing
they don't know what kind of cell they
are
and they
don't know where they are
and then
they sharpen
their expectations about what kind of
cell they are
while also
moving into a different spatial niche
but they're not tracking their location
like i'm at three comma two where are
you
it's like this is what i'm biochemically
sensing
and so it's a relational morphology
that doesn't need the blueprint in the
nucleus
so it's a lot like an ant colony there's
not the nest architecture blueprint in
the brain there's the process for stigma
g
and then they show that with some other
um changes
in
the generative process
a positive squared gradient in the
generative process
they get double head formation
which some modulations have been shown
to empirically result in in the lab and

they can also make it so it gets double
tail formation

so it's just recapitulating this basic
example and then showing that like
modifying the external field changes the
morphology that gets attracted to
so those are the key pieces that's
modeling morphogenesis

as

bayesian inference

reducing surprise

with a variational free energy flow on
action and internal states action states
internal states

okay

um

we're not gonna go into it here
definitely for the dot one and dot two
so um 3.1 is the construction of the
model

they then

unpack that signal

matrix

the combination of the signals

and then um

[Music]

they give even more information about
modeling signals

and they also imbue it here with like
this steminess

which helps the cells start out with um

i believe a weaker prior

about what kind of cell they are which
translates to where they should be

um

in figure five

they do a targeted intervention in their
studying of anomalous cell behavior

and so basically they start off with these um cells that have initially unspecified state and they converge in a this panels on a by 32 time points they're converged to the target morphology

but in b

they show that if one of the cells that gets hit with a white arrow here has a perturbed signaling response it fails to correctly infer its place in the ensemble

so b is like sort of like um it could be like a genetic mutation or a targeted modification of the signaling and it's like that third red cell

never forms

and then in c

the same aberrant cell from b is initially is rescued by an increased signaling sensitivity of other cells leading another cell

green arrow

to switch position with the aberrant cell pink arrow

pretty

cool okay yeah i wanted to come up with an analogy that would actually work that would explain this like if

you're out geocaching one afternoon and then all of a sudden

some

something was able to turn all the lights off and then the person that was geocaching beside you change their behavior but i couldn't even come up with a valid analogy to try to explain this

how about um

uh we're playing american football

and so there's a plan and everyone's

trying to reduce their uncertainty about

how the plan is playing out from their

perspective

and so if all goes to plan we're all

going to deploy into the exact right

positions

but then

somebody um doesn't move

if each nestmate on the football team

has low sensitivity

then they're going to continue in their

own expected trajectory as if nothing

had happened that's kind of like what we

see in b it's like a partial rollout

whereas

this is changing the environment mixing

our metaphors etc

so that a high sensitivity

teammate

fills in for that critical position so

that the attractor state of the strategy

can still be morphologically realized

even though

um there still is that one individual

who's not moving

but but they're not interfering

with the strategy's attraction

but we'll come up with probably some

other and better ways to talk about it

okay

and then they um just mentioned that

this is something that can be modeled

and it's in spm

and also on franz kuchling's um

github

so maybe we will um look at the code
and then

just to say that we're not going to go
into it at all today but all of section
4 is super interesting

they review some of the mathematical
assumptions and limitations of the model
there's a really fascinating discussion
about variational principles and open
systems

and so um

[Music]

like just to read we have shown that the
variational free energy minimization in
active inference is related to the
variational principle of least action
it is worth pointing out with where
these two approaches
diverge

two instruments

in a wood diverged

and then it's all about that divergence
and the contrasting

and so

i think there's gonna be a lot of
unpacked there

and then they have some closing remarks
on the applicability in biological
systems and some of the predictive
capacities of the simulations

um in terms of like you could make a
simulation of a healthy functioning
tissue and then have predictive capacity
about asking counterfactuals well what
if this happened what might i expect so
that is

a lot of info so

thanks to everybody who's been

watching and

again hope that especially with some of these technical parts

that we were able to represent it with high integrity

so you won't have the last word but what would you say in closing as we move from the dot zero into the one two and beyond well first of all you went through a heck of a lot of stuff in a very short period of time so that's that's quite the gradient flow right there so that's impressive

um

i don't think much in terms of i mean this is a classic example of where you have to unpack

as opposed to

lots of the other papers that are maybe more philosophically based whether you you

provision like you pack a bunch of stuff and then you kind of go off on a bit of a journey and i think in the point one and the point two we may still be doing a little bit more unpacking just because of the

the the density of this kind of information but that doesn't i mean you want to be able to do both so i think i think being able to cover

this and

hopefully we got it right like i

i did my best to try to understand it

and i think he did a good job of

explaining it so we'll see who shows up in the point one and two whether the point one and two or more unpacking or

maybe

starting to actually

think of ways that we could employ this

you already mentioned that we could we

can apply it to digital teams but

whether or not people have the

confidence to be able to take this and

apply it like they've done to

morphogenesis

maybe that's part of that conversation

cool

yeah i'll be looking forward to

going through with multiple

concurrent regimes of attention

some of the formalisms that we either um

disgraced or skipped in this discussion

and then keeping it open

to think about

morphogenesis like

where has morphogenesis been in numbers

1 through 38

why haven't we been talking about

morphogenesis and if not morphogenesis

then what else have we been focusing on

um

[Music]

how does it relate to embodiment

and to spatial and physical aspects of

cognition

there's so many interesting angles

and uh

i'm sure we'll have a lot to discuss so

dean thanks a ton for all the help on

the slides and for this discussion

um

[Music]

hope to see you in the coming weeks

all right thanks daniel take care peace

see you later